An AI-Based Document Translation System with Original Format Preservation
'n KI-gebaseerde dokumentvertaalstelsel met oorspronklike formaatbewaring

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Abstract. In the context of the increasing demand for digital document translation, modern translation systems are expected not only to ensure the quality of translated content but also to preserve the original formatting of input documents. However, when translating structurally complex documents such as PDF and DOCX files, many systems still encounter problems such as layout distortion, table misalignment, image displacement, font errors, and limited support for bilingual display modes. This study proposes an artificial intelligence-based document translation system capable of processing two common document formats, DOCX and PDF, while supporting three translation modes: standard translation, side-by-side bilingual translation, and line-by-line bilingual translation. For DOCX documents, the system directly processes document components such as paragraphs, tables, headers, and footers. For PDF documents, the system applies a PDF→DOCX→translation→layout recovery→PDF pipeline to improve format preservation. The system integrates several translation providers, including Google Translate, DeepL Translator, and Gemini 2.5 Flash, to increase flexibility in translation quality and processing options. The experimental evaluation is conducted using two groups of criteria: formatting preservation and translation quality measured by BLEU Mean and BLEU Std. The results show that the proposed system achieves strong format preservation for DOCX documents and promising performance for academic PDF documents.

Abstrak. In die konteks van die toenemende vraag na digitale dokumentvertaling, word van moderne vertaalstelsels verwag om nie net die kwaliteit van vertaalde inhoud te verseker nie, maar ook om die oorspronklike formatering van insetdokumente te bewaar. Wanneer struktureel komplekse dokumente soos PDF- en DOCX-lêers vertaal word, ondervind baie stelsels egter steeds probleme soos uitlegvervorming, tabelwanbelyning, beeldverplasing, fontfoute en beperkte ondersteuning vir tweetalige vertoonmodusse. Hierdie studie stel 'n kunsmatige intelligensie-gebaseerde dokumentvertalingstelsel voor wat in staat is om twee algemene dokumentformate, DOCX en PDF te verwerk, terwyl dit drie vertaalmodusse ondersteun: standaardvertaling, tweetalige vertaling langs mekaar en tweetalige lyn-vir-lyn vertaling. Vir DOCX-dokumente verwerk die stelsel dokumentkomponente soos paragrawe, tabelle, kop- en voettekste direk. Vir PDF-dokumente pas die stelsel 'n PDF→DOCX→vertaling→uitlegherstel→PDF-pyplyn toe om formaatbewaring te verbeter. Die stelsel integreer verskeie vertaalverskaffers, insluitend Google Translate, DeepL Translator en Gemini 2.5 Flash, om buigsaamheid in vertalingskwaliteit en verwerkingsopsies te verhoog. Die eksperimentele evaluering word uitgevoer deur gebruik te maak van twee groepe kriteria: formatering bewaring en vertaling kwaliteit gemeet deur BLEU Mean en BLEU Std. Die resultate toon dat die voorgestelde stelsel sterk formaatbewaring vir DOCX-dokumente behaal en belowende prestasie vir akademiese PDF-dokumente.

Keywords: Document translation, format preservation, artificial intelligence, PDF, DOCX, bilingual translation, BLEU.
Sleutelwoorde: Dokumentvertaling, formaatbewaring, kunsmatige intelligensie, PDF, DOCX, tweetalige vertaling, BLEU.

1 Introduction
1 Inleiding

The rapid development of artificial intelligence and large language models has significantly advanced the field of natural language processing, especially machine translation. Modern translation models are no longer limited to converting text from a source language to a target language; they are also capable of handling context, terminology, and writing style more effectively than traditional statistical translation approaches(Bahdanau et al., 2015; Wu et al., 2016; Koehn, 2020). In particular, the Transformer architecture has become a fundamental foundation for many modern

Die vinnige ontwikkeling van kunsmatige intelligensie en groot taalmodelle het die veld van natuurlike taalverwerking, veral masjienvertaling, aansienlik bevorder. Moderne vertaalmodelle is nie meer beperk tot die omskakeling van teks van 'n brontaal na 'n doeltaal nie; hulle is ook in staat om konteks, terminologie en skryfstyl meer effektief te hanteer as tradisionele statistiese vertaalbenaderings (Bahdanau et al., 2015; Wu et al., 2016; Koehn, 2020). In die besonder het die Transformer-argitektuur 'n fundamentele grondslag vir baie moderne geword language processing models due to its ability to capture contextual relationships efficiently in long text sequences (Vaswani et al., 2017). In addition, the development of large language models such as GPT and unified text-to-text models has opened new directions for translation, summarization, and document processing tasks (Brown et al., 2020; Raffel et al., 2020). However, in practical scenarios, users do not only need to translate plain sentences or paragraphs. They often need to translate entire formatted documents such as scientific papers, technical reports, theses, contracts, and administrative forms. These documents are commonly stored in DOCX or PDF formats and contain various formatting components, including headings, paragraphs, tables, images, headers, footers, lists, and page layouts. Therefore, document translation should not be considered only as a content translation problem, but also as a formatting preservation problem.

taalverwerkingsmodelle vanweë die vermoë daarvan om kontekstuele verhoudings doeltreffend in lang teksreekse vas te vang (Vaswani et al., 2017). Daarbenewens het die ontwikkeling van groot taalmodelle soos GPT en verenigde teks-tot-teks-modelle nuwe rigtings vir vertaal-, opsomming- en dokumentverwerkingstake geopen (Brown et al., 2020; Raffel et al., 2020). In praktiese scenario's hoef gebruikers egter nie net gewone sinne of paragrawe te vertaal nie. Hulle moet dikwels volledige geformateerde dokumente vertaal, soos wetenskaplike referate, tegniese verslae, tesisse, kontrakte en administratiewe vorms. Hierdie dokumente word gewoonlik in DOCX- of PDF-formate gestoor en bevat verskeie formateringskomponente, insluitend opskrifte, paragrawe, tabelle, prente, kopskrifte, voettekste, lyste en bladsyuitlegte. Dokumentvertaling moet dus nie net as 'n inhoudsvertalingsprobleem beskou word nie, maar ook as 'n formateringsbehoudprobleem.

In many existing document translation systems, preserving the original format remains a considerable challenge. DOCX has a relatively structured representation in which content is organized into paragraphs, tables, styles, headers, and footers, making it possible to process the document directly using programming tools. In contrast, PDF is mainly designed for display and printing, and its content is usually stored based on page coordinates rather than a logical document structure. As a result, when translated text becomes longer or shorter than the original text, problems such as text overflow, table misalignment, overlapping text, and layout changes may occur. Popular translation services such as Google Translate, DeepL, Adobe Acrobat, and AI platforms with file-upload support provide convenient solutions, but they still have limitations in terms of pipeline customization, bilingual presentation, and output format control. Based on these limitations, this study develops an AI-based document translation system that preserves the original document format and supports both DOCX and PDF files with three translation modes. The system uses different processing strategies for different document types: DOCX files are translated directly based on their structured paragraphs and tables, while PDF files are processed using a PDF→DOCX→translation→layout recovery→PDF pipeline. The proposed system is evaluated through two experimental groups: formatting preservation evaluation and translation quality evaluation, in order to assess both visual structure and linguistic content.

In baie bestaande dokumentvertalingstelsels bly die behoud van die oorspronklike formaat 'n aansienlike uitdaging. DOCX het 'n relatief gestruktureerde voorstelling waarin inhoud in paragrawe, tabelle, style, kop- en voettekste georganiseer word, wat dit moontlik maak om die dokument direk met behulp van programmeringsinstrumente te verwerk. Daarteenoor is PDF hoofsaaklik ontwerp vir vertoon en druk, en die inhoud daarvan word gewoonlik geberg op grond van bladsykoördinate eerder as 'n logiese dokumentstruktuur. As gevolg hiervan, wanneer vertaalde teks langer of korter as die oorspronklike teks word, kan probleme soos teksoorloop, tabelwanbelyning, oorvleuelende teks en uitlegveranderinge voorkom.

Gewilde vertaaldienste soos Google Translate, DeepL, Adobe Acrobat en KI-platforms met lêeroplaai-ondersteuning bied gerieflike oplossings, maar hulle het steeds beperkings in terme van pyplynaanpassing, tweetalige aanbieding en uitvoerformaatbeheer. Gebaseer op hierdie beperkings, ontwikkel hierdie studie 'n KI-gebaseerde dokumentvertalingstelsel wat die oorspronklike dokumentformaat bewaar en beide DOCX- en PDF-lêers met drie vertaalmodusse ondersteun. Die stelsel gebruik verskillende verwerkingstrategieë vir verskillende dokumenttipes: DOCX-lêers word direk vertaal op grond van hul gestruktureerde paragrawe en tabelle, terwyl PDF-lêers verwerk word deur 'n PDF→DOCX→vertaling→uitlegherstel→PDF-pyplyn te gebruik. Die voorgestelde stelsel word deur twee eksperimentele groepe geëvalueer: formateringsbewaring-evaluering en vertalingskwaliteit-evaluering, ten einde beide visuele struktuur en linguistiese inhoud te assesseer.

2 Related Work
2 Verwante werk

In the field of document translation, many commercial systems have been developed to help users translate common file formats such as DOCX, PDF, and PPTX. Platforms such as DeepL Document Translation, Google Translate, Google Cloud Translation, and Adobe Acrobat provide online document translation features, allowing users to process multilingual documents for study, research, and professional work. These systems offer advantages in terms of processing speed, translation quality, and usability. However, most commercial platforms are closed systems, meaning that users cannot easily modify the internal processing pipeline, customize format preservation mechanisms, or flexibly control bilingual presentation modes such as side-by-side bilingual translation or line-by-line bilingual translation. For users working with academic or technical documents that contain complex structures,
Op die gebied van dokumentvertaling is baie kommersiële stelsels ontwikkel om gebruikers te help om algemene lêerformate soos DOCX, PDF en PPTX te vertaal. Platforms soos DeepL Document Translation, Google Translate, Google Cloud Translation en Adobe Acrobat bied aanlyn dokumentvertalingskenmerke, wat gebruikers in staat stel om meertalige dokumente vir studie, navorsing en professionele werk te verwerk. Hierdie stelsels bied voordele in terme van verwerkingspoed, vertaalkwaliteit en bruikbaarheid. Die meeste kommersiële platforms is egter geslote stelsels, wat beteken dat gebruikers nie maklik die interne verwerkingspyplyn kan verander, formaatbewaringsmeganismes kan aanpas of tweetalige aanbiedingsmodusse soos sy-aan-sy tweetalige vertaling of lyn-vir-lyn tweetalige vertaling buigsaam kan beheer nie. Vir gebruikers wat met akademiese of tegniese dokumente werk wat komplekse strukture bevat,

relying only on existing commercial file translation tools may not be sufficient to control formatting errors generated during the translation process.

om slegs op bestaande kommersiële lêervertaalinstrumente te vertrou, is dalk nie voldoende om formateringsfoute wat tydens die vertaalproses gegenereer word, te beheer nie.

From an academic perspective, machine translation has evolved significantly from statistical machine translation to neural machine translation and large language models. The attention mechanism in neural machine translation improved the ability to align source and target sentences(Bahdanau et al., 2015), while Google Neural Machine Translation demonstrated the effectiveness of neural networks in large-scale multilingual translation (Wu et al., 2016). Later, the Transformer architecture became a key foundation for modern translation and natural language processing models through its self-attention mechanism(Vaswani et al., 2017). Reviews of neural machine translation also indicate that translation quality has improved considerably, although challenges remain in context handling, domain-specific terminology, and evaluation methodology(Stahlberg, 2020; Koehn, 2020). In addition, large language models such as GPT and unified text-to-text models have expanded the ability to process long text, understand context, and generate more natural outputs (Brown et al., 2020; Raffel et al., 2020).

Vanuit 'n akademiese perspektief het masjienvertaling aansienlik ontwikkel van statistiese masjienvertaling na neurale masjienvertaling en groot taalmodelle. Die aandagmeganisme in neurale masjienvertaling het die vermoë verbeter om bron- en teikensinne in lyn te bring (Bahdanau et al., 2015), terwyl Google Neurale Masjienvertaling die doeltreffendheid van neurale netwerke in grootskaalse veeltalige vertaling gedemonstreer het (Wu et al., 2016). Later het die Transformer-argitektuur 'n sleutelgrondslag vir moderne vertalings- en natuurlike taalverwerkingsmodelle geword deur sy selfaandagmeganisme (Vaswani et al., 2017). Resensies van neurale masjienvertaling dui ook aan dat vertalingskwaliteit aansienlik verbeter het, alhoewel daar nog uitdagings is in kontekshantering, domeinspesifieke terminologie en evalueringsmetodologie (Stahlberg, 2020; Koehn, 2020). Boonop het groot taalmodelle soos GPT en verenigde teks-tot-teks-modelle die vermoë uitgebrei om lang teks te verwerk, konteks te verstaan en meer natuurlike uitsette te genereer (Brown et al., 2020; Raffel et al., 2020).

For document processing, DOCX and PDF formats have very different structural characteristics. DOCX is based on an XML structure, where content is organized into paragraphs, text runs, tables, styles, headers, and footers. This allows a system to iterate through document components, extract and replace text, and preserve most of the original formatting. In contrast, PDF is designed mainly for display and printing, and its text is often stored according to page coordinates. This makes reading-order detection, table recognition, and layout recovery more difficult. Studies on document layout analysis, such as PubLayNet, DocBank, FUNSD, DocVQA, and LayoutLM, show that understanding the visual and structural layout of documents is important for complex document processing tasks(Zhong et al., 2019; Li et al., 2020; Jaume et al., 2019; Mathew et al., 2021; Xu et al., 2020). In particular, LayoutLM and LayoutLMv2 combine textual, positional, and visual information to improve document understanding for visually rich documents (Xu et al., 2020; Xu et al., 2021).

Vir dokumentverwerking het DOCX- en PDF-formate baie verskillende strukturele eienskappe. DOCX is gebaseer op 'n XML-struktuur, waar inhoud in paragrawe, tekslopies, tabelle, style, kop- en voettekste georganiseer word. Dit laat 'n stelsel toe om deur dokumentkomponente te herhaal, teks te onttrek en te vervang, en die meeste van die oorspronklike formatering te bewaar. Daarteenoor is PDF hoofsaaklik vir vertoon en druk ontwerp, en die teks word dikwels volgens bladsykoördinate gestoor. Dit maak leesvolgorde opsporing, tabelherkenning en uitlegherwinning moeiliker. Studies oor dokumentuitlegontleding, soos PubLayNet, DocBank, FUNSD, DocVQA en LayoutLM, toon dat die begrip van die visuele en strukturele uitleg van dokumente belangrik is vir komplekse dokumentverwerkingstake (Zhong et al., 2019; Li et al., 2020; Jaume et al., 2019; Mathew et al., 2019; Mathew et al., X20. 2020). In die besonder kombineer LayoutLM en LayoutLMv2 tekstuele, posisionele en visuele inligting om dokumentbegrip vir visueel ryk dokumente te verbeter (Xu et al., 2020; Xu et al., 2021).

A practical approach for PDF processing is to convert PDF files into an editable format such as DOCX, translate the extracted content, and then export the result back to PDF. The PDF→DOCX→translation→PDF pipeline has the advantage of using the flexible editing structure of Word documents, allowing the system to process text by paragraphs, tables, and styles instead of manipulating raw PDF coordinates directly. However, this pipeline also has limitations because the conversion process may introduce intermediate errors such as merged text, fragmented paragraphs, incorrectly detected tables, or modified page layouts. Therefore, for this pipeline to work effectively, preprocessing, DOCX sanitization, layout recovery, and output quality checking are required. For scanned documents or embedded images, OCR is also important for creating a text layer that can be processed, and Tesseract is one of the most widely used OCR engines in both research and practical applications (Smith, 2007).

'n Praktiese benadering vir PDF-verwerking is om PDF-lêers in 'n redigeerbare formaat soos DOCX te omskep, die onttrekte inhoud te vertaal en dan die resultaat terug na PDF uit te voer. Die PDF→DOCX→vertaling→PDF-pyplyn het die voordeel om die buigsame redigeringstruktuur van Word-dokumente te gebruik, wat die stelsel toelaat om teks volgens paragrawe, tabelle en style te verwerk in plaas daarvan om rou PDF-koördinate direk te manipuleer. Hierdie pyplyn het egter ook beperkings omdat die omskakelingsproses intermediêre foute soos saamgevoegde teks, gefragmenteerde paragrawe, verkeerde bespeurde tabelle of gewysigde bladsyuitlegte kan inbring. Daarom, vir hierdie pyplyn om doeltreffend te werk, word voorafverwerking, DOCX-sanering, uitlegherwinning en uitsetkwaliteitkontrolering vereis. Vir geskandeerde dokumente of ingebedde beelde is OCR ook belangrik vir die skep van 'n tekslaag wat verwerk kan word, en Tesseract is een van die mees gebruikte OCR-enjins in beide navorsing en praktiese toepassings (Smith, 2007).

Machine translation evaluation studies often use automatic metrics such as BLEU, TER, METEOR, and COMET to measure similarity between machine translations and reference translations(Papineni et al., 2002; Snover et al., 2006; Banerjee& Lavie, 2005; Rei et al., 2020). Among them, BLEU is a widely used metric based on n-gram overlap between a system-generated translation and a reference translation

Masjienvertalingsevalueringstudies gebruik dikwels outomatiese maatstawwe soos BLEU, TER, METEOR en COMET om ooreenkomste tussen masjienvertalings en verwysingsvertalings te meet (Papineni et al., 2002; Snover et al., 2006; Banerjee & Lavie, 2005; Rei et al., 2020). Onder hulle is BLEU 'n wyd gebruikte maatstaf gebaseer op n-gram oorvleueling tussen 'n stelselgegenereerde vertaling en 'n verwysingsvertaling

(Papineni et al., 2002). However, BLEU scores must be reported clearly because they can vary depending on tokenization, normalization, and calculation settings(Post, 2018). For document translation with format preservation, translation quality alone is not sufficient. A system may translate content correctly but still fail in practical use if it loses tables, shifts images, distorts layout, or produces an unreadable output document. Therefore, this study evaluates the proposed system from two perspectives: formatting preservation and translation quality.

(Papineni et al., 2002). BLEU-tellings moet egter duidelik gerapporteer word, want dit kan wissel na gelang van tokenisering, normalisering en berekeninginstellings (Post, 2018). Vir dokumentvertaling met formaatbehoud is vertaalkwaliteit alleen nie voldoende nie. 'n Stelsel kan inhoud korrek vertaal, maar steeds misluk in praktiese gebruik as dit tabelle verloor, beelde verskuif, uitleg verdraai of 'n onleesbare uitvoerdokument produseer. Daarom evalueer hierdie studie die voorgestelde stelsel vanuit twee perspektiewe: formatering bewaring en vertaling kwaliteit.

Based on the above analysis, the research gap lies in combining three factors: translation quality, original format preservation, and flexible bilingual presentation modes. Existing systems usually focus on either content translation or format preservation to a certain extent, but they do not fully support standard translation, side-by-side bilingual translation, and line-by-line bilingual translation within a customizable and extensible pipeline. Furthermore, allowing users to choose among translation providers such as Google Translate, DeepL, and Gemini makes the system more adaptable to different requirements in cost, speed, and translation quality. Therefore, this study proposes an AI-based document translation system that combines DOCX processing, a PDF→DOCX→translation→PDF pipeline, special-content protection, layout recovery, and experimental evaluation of both formatting and translation quality.

Op grond van bogenoemde ontleding lê die navorsingsgaping in die kombinasie van drie faktore: vertalingskwaliteit, oorspronklike formaatbewaring en buigsame tweetalige aanbiedingswyses. Bestaande stelsels fokus gewoonlik tot 'n sekere mate op óf inhoudvertaling óf formaatbewaring, maar hulle ondersteun nie standaardvertaling, sy-aan-sy tweetalige vertaling en lyn-vir-lyn tweetalige vertaling binne 'n aanpasbare en uitbreidbare pyplyn nie. Verder, deur gebruikers toe te laat om tussen vertaalverskaffers soos Google Translate, DeepL en Gemini te kies, maak die stelsel meer aanpasbaar by verskillende vereistes in koste, spoed en vertaalkwaliteit. Daarom stel hierdie studie 'n KI-gebaseerde dokumentvertalingstelsel voor wat DOCX-verwerking, 'n PDF→DOCX→vertaling→PDF-pyplyn, spesiale-inhoudbeskerming, uitlegherwinning en eksperimentele evaluering van beide formatering en vertaalkwaliteit kombineer.

3 System Design
3 Stelselontwerp

The proposed system is designed to solve the problem of translating documents while preserving the original format for two main document types: DOCX and PDF. The overall workflow begins when a user uploads a document through the web interface and selects the target language, translation provider, and translation mode. After receiving the request, the system validates the file format, stores the input document, and creates a background processing job. This asynchronous processing design is suitable for large documents or documents that require multiple intermediate processing steps, especially PDF files. During translation, the user interface can monitor the processing status through a job identifier and allow the user to download the output file when the job is completed.

Die voorgestelde stelsel is ontwerp om die probleem van die vertaling van dokumente op te los, terwyl die oorspronklike formaat vir twee hoofdokumenttipes bewaar word: DOCX en PDF. Die algehele werkvloei begin wanneer 'n gebruiker 'n dokument deur die webkoppelvlak oplaai en die teikentaal, vertaalverskaffer en vertaalmodus kies. Nadat die versoek ontvang is, bevestig die stelsel die lêerformaat, stoor die invoerdokument en skep 'n agtergrondverwerkingstaak. Hierdie asinchroniese verwerkingsontwerp is geskik vir groot dokumente of dokumente wat veelvuldige intermediêre verwerkingstappe vereis, veral PDF-lêers. Tydens vertaling kan die gebruikerskoppelvlak die verwerkingstatus monitor deur 'n posidentifiseerder en die gebruiker toelaat om die uitvoerlêer af te laai wanneer die taak voltooi is.

The system architecture consists of several main components: the user interface, the request-handling API, the background job manager, the file processing service, the translation provider module, the layout recovery module, and the output generation module. The user interface allows users to upload DOCX or PDF files, select the target language, choose a translation provider such as Google Translate, DeepL, or Gemini, and select one of three translation modes: standard translation, side-by-side bilingual translation, and line-by-line bilingual translation. The API validates the uploaded file, stores it, creates a translation record, and initializes a background job. The file processing service then determines the document type and selects the appropriate pipeline. This architecture allows the system to be extended with additional translation providers or new document formats in the future.

Die stelselargitektuur bestaan uit verskeie hoofkomponente: die gebruikerskoppelvlak, die versoekhantering-API, die agtergrondwerkbestuurder, die lêerverwerkingsdiens, die vertaalverskaffermodule, die uitlegherwinningsmodule en die uitsetgenereringsmodule. Die gebruikerskoppelvlak stel gebruikers in staat om DOCX- of PDF-lêers op te laai, die teikentaal te kies, 'n vertaalverskaffer soos Google Translate, DeepL of Gemini te kies, en een van drie vertaalmodusse te kies: standaardvertaling, tweetalige vertaling langs mekaar en tweetalige lyn-vir-lyn vertaling. Die API bekragtig die opgelaaide lêer, stoor dit, skep 'n vertaalrekord en inisialiseer 'n agtergrondtaak. Die lêerverwerkingsdiens bepaal dan die dokumenttipe en kies die toepaslike pyplyn. Hierdie argitektuur laat toe dat die stelsel in die toekoms uitgebrei kan word met bykomende vertaalverskaffers of nuwe dokumentformate.

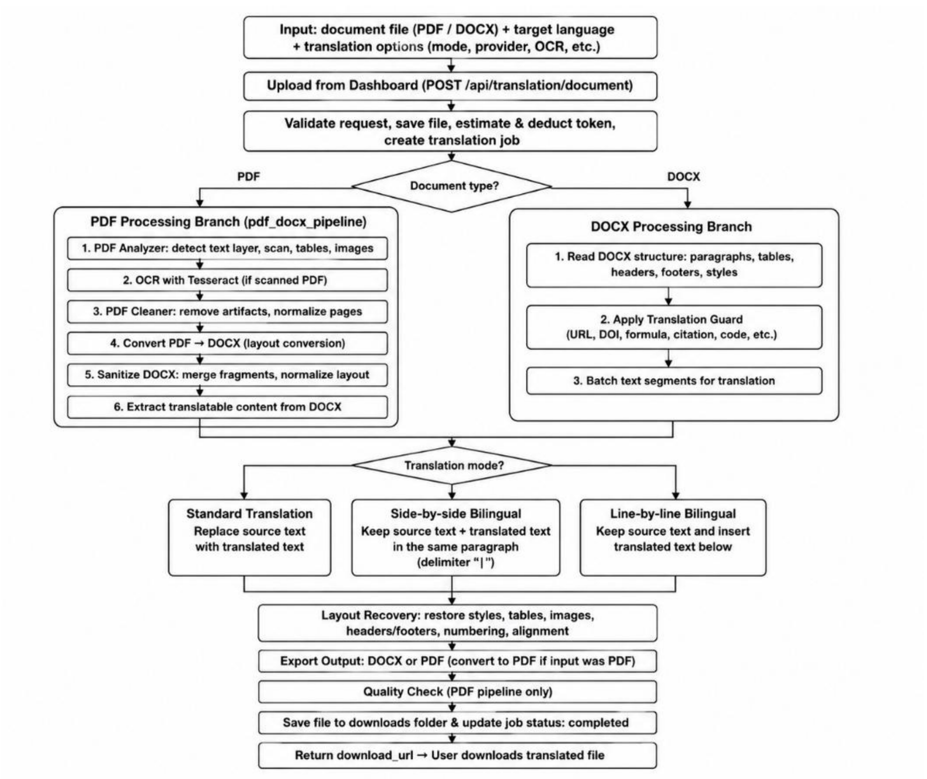

Fig. 1.*Workflow of the proposed AI-based document translation system with original format preservation.*

Fig. 1. Werkvloei van die voorgestelde KI-gebaseerde dokumentvertalingstelsel met oorspronklike formaatbewaring.

For DOCX documents, the system directly processes the document structure by iterating through paragraphs, tables, headers, and footers. Since DOCX has a clear internal structure, the system can preserve most styles, fonts, margins, and table structures while modifying only textual content. Before sending text to a translation provider, the system applies a Translation Guard mechanism to protect non-translatable elements such as URLs, DOIs, email addresses, citations, formulas, technical symbols, and code snippets. This mechanism is especially important for academic and technical documents because translating such elements incorrectly may change their original meaning. After translation, the translated text is written back into the document according to the selected translation mode.

Vir DOCX-dokumente verwerk die stelsel die dokumentstruktuur direk deur paragrawe, tabelle, kopskrifte en voettekste te herhaal. Aangesien DOCX 'n duidelike interne struktuur het, kan die stelsel die meeste style, lettertipes, kantlyne en tabelstrukture bewaar terwyl slegs tekstuele inhoud gewysig word. Voordat teks aan 'n vertaalverskaffer gestuur word, pas die stelsel 'n Translation Guard-meganisme toe om nie-vertaalbare elemente soos URL's, DOI's, e-posadresse, aanhalings, formules, tegniese simbole en kodebrokkies te beskerm. Hierdie meganisme is veral belangrik vir akademiese en tegniese dokumente omdat die verkeerde vertaling van sulke elemente hul oorspronklike betekenis kan verander. Na vertaling word die vertaalde teks teruggeskryf in die dokument volgens die gekose vertaalmodus.

For PDF documents, the system uses an intermediate conversion pipeline: PDF→DOCX→translation→layout recovery→PDF. First, the PDF file is analyzed to determine whether it contains a text layer, whether it is a scanned document, and whether it includes tables, images, or complex page layouts. If the file is scanned or contains image-based text, OCR can be applied to create a processable text layer. The PDF is then cleaned and converted into DOCX. Since PDF-to-DOCX conversion may create issues such as fragmented paragraphs, merged text, incorrect table detection, or abnormal line spacing, the system performs a sanitization step to merge fragmented

Vir PDF-dokumente gebruik die stelsel 'n intermediêre omskakelingspyplyn: PDF→DOCX→vertaling→uitlegherstel→PDF. Eerstens word die PDF-lêer ontleed om te bepaal of dit 'n tekslaag bevat, of dit 'n geskandeerde dokument is, en of dit tabelle, prente of komplekse bladsyuitlegte insluit. As die lêer geskandeer is of beeldgebaseerde teks bevat, kan OCR toegepas word om 'n verwerkbare tekslaag te skep. Die PDF word dan skoongemaak en in DOCX omgeskakel. Aangesien PDF-na-DOCX-omskakeling probleme kan veroorsaak soos gefragmenteerde paragrawe, saamgevoegde teks, verkeerde tabelbespeuring of abnormale reëlspasiëring, voer die stelsel 'n ontsmettingsstap uit om gefragmenteerde saam te voeg segments, normalize the layout, and reduce noise before translation. After translation, the Layout Recovery module is used to restore the document layout before exporting the result back to PDF.

segmente, normaliseer die uitleg en verminder geraas voor vertaling. Na vertaling word die uitlegherstelmodule gebruik om die dokumentuitleg te herstel voordat die resultaat terug na PDF uitgevoer word.

The system supports three translation modes to meet different user needs. In standard translation mode, the original text is replaced by the translated text, which is suitable when the user only needs the final document in the target language. In side-by-side bilingual translation mode, the system keeps the original text and places the translated text in the same paragraph, usually separated by the"|" delimiter, allowing users to quickly compare the source and target text. In line-by-line bilingual translation mode, the system keeps the original paragraph and inserts the translated paragraph immediately below it. This mode is useful for language learning, academic reading, and translation checking. Since bilingual modes increase the length of the document, the goal of layout recovery is not to preserve the exact number of pages, but to ensure that the content remains readable, correctly ordered, complete, and free from serious text-overlapping errors.

Die stelsel ondersteun drie vertaalmodusse om aan verskillende gebruikersbehoeftes te voldoen. In standaardvertaalmodus word die oorspronklike teks vervang deur die vertaalde teks, wat geskik is wanneer die gebruiker slegs die finale dokument in die doeltaal benodig. In sy-aan-sy tweetalige vertaalmodus hou die stelsel die oorspronklike teks en plaas die vertaalde teks in dieselfde paragraaf, gewoonlik geskei deur die "|" skeidingsteken, wat gebruikers in staat stel om die bron- en teikenteks vinnig te vergelyk. In reël-vir-reël tweetalige vertaalmodus, behou die stelsel die oorspronklike paragraaf en voeg die vertaalde paragraaf onmiddellik daaronder in. Hierdie modus is nuttig vir taalleer, akademiese lees en vertalingskontrole. Aangesien tweetalige modusse die lengte van die dokument vergroot, is die doel van uitlegherwinning nie om die presiese aantal bladsye te bewaar nie, maar om te verseker dat die inhoud leesbaar, korrek georden, volledig en vry van ernstige teksoorvleuelende foute bly.

Algorithm 1
Input:
F: input document (. pdf or . docx)
L: target language
P: translation provider (Google Translate, DeepL, Gemini)
M: translation mode (standard, side-by-side bilingual, line-by-line bilingual)
Opt: optional settings such as OCR and delimiter configuration

Algoritme 1
Invoer:
F: invoerdokument (. pdf of . docx)
L: doeltaal
P: vertaalverskaffer (Google Translate, DeepL, Gemini)
M: vertaalmodus (standaard, sy-aan-sy tweetalig, reël-vir-reël tweetalig)
Opt: opsionele instellings soos OCR en skeidingskonfigurasie

Algoritme 1
Invoer:

```
   F: invoerdokument (. pdf van . docx)
   L: doeltaal
   P: vertaalverskaffer (Google Translate, DeepL,
Gemini)
   M: vertaalmodus (standaard, sy-aan-sy tweetalig,
reël-vir-reël tweetalig)
Opt: opsionele instellings soos OCR en
skeidingskonfigurasie

   Output:
      F_out: translated document with preserved
   formatting

   Uitset:
       F_out: vertaalde dokument met
   bewaarde formatering

   BEGIN
       Validate the input document F and target language L

BEGIN
   Valideer die invoerdokument F en doeltaal L

      IF the file extension of F is not . pdf or .
         docx THEN      Return unsupported
         file format error
         END IF

   AS die lêeruitbreiding van F nie . pdf of . docx DAN
      Gee ongesteunde lêerformaatfout terug
   EINDE INDIEN

   AS dies leeruitbreiding van F nie . pdf van. docx
   DAN          Gee ongesteunde peerformaat terug
   EINDE INDIEN

   Save F to upload storage
   Create a background translation job
   Return job_id and status_url to the client

   Stoor F om berging op te laai
   Skep 'n agtergrondvertaalwerk
   Gee job_id en status_url terug na die
   kliënt

   Set the job status to in_progress

   Stel die werkstatus op in_progress

   IF F is a PDF document THEN
      Analyze the PDF structure, including text layer, scanned pages, tables, and
images

AS F 'n PDF-dokument DAN is
      Ontleed die PDF-struktuur, insluitend tekslaag, geskandeerde bladsye,
tabelle en beelde

      IF the PDF is scanned THEN
         Apply OCR to create a searchable PDF
         IF OCR fails in strict mode THEN
```

AS die PDF DAN geskandeer word
Pas OCR toe om 'n soekbare PDF te
skep AS OCR DAN in streng modus
misluk

```
          Set the job status to failed
          Return error
        END IF
      END IF

      Stel die taakstatus op misluk
              Terugkeer fout
       EINDE INDIEN
      EINDE INDIEN

Clean the PDF document
Convert the PDF document to DOCX Sanitize the converted DOCX document

      Maak die PDF-dokument skoon
      Skakel die PDF-dokument om na DOCX
      Maak die omgeskakelde DOCX-
      dokument skoon

       Extract paragraphs, tables, headers, and footers from
      the DOCX document          Apply translation guard to
      protect URLs, DOIs, citations, formulas, and  special
      symbols
       Translate the extracted content using provider P and
      mode M
       Recover the document layout after translation

      Onttrek paragrawe, tabelle, kop- en voettekste uit die
    DOCX-dokument     Pas vertaalwag toe om URL's, DOI's,
    aanhalings, formules en spesiale simbole te beskerm
      Vertaal die onttrekte inhoud deur verskaffer P en modus M
    te gebruik
      Herwin die dokumentuitleg na vertaling

    IF M is standard translation THEN
       Apply the final layout refinement step END IF

    AS M standaardvertaling DAN is
      Pas die finale uitlegverfyningstap EINDE IF toe

Convert the translated DOCX document
back to PDF  Perform quality checking on
the output PDF

      Skakel die vertaalde DOCX-dokument terug na PDF
      Voer kwaliteitskontrole uit op die uitvoer PDF

    ELSE IF F is a DOCX document THEN
      Open the DOCX document directly
      Extract paragraphs, tables, headers, and footers
      Apply translation guard to protect special content
      Translate the DOCX content using provider P and mode M
      Preserve the original styles and document structure
    END IF
```

```
    ANDERS AS F DAN 'n DOCX-dokument is
        Maak die DOCX-dokument direk oop
        Onttrek paragrawe, tabelle, koptekste en voettekste
        Pas vertaalwag toe om spesiale inhoud te beskerm
        Vertaal die DOCX-inhoud deur verskaffer P en modus M
    te gebruik   Behou die oorspronklike style en
    dokumentstruktuur
    EINDE INDIEN

    Save the translated document to download storage
    Set the job status to completed
    Return the download path of F_out

    Stoor die vertaalde dokument om berging af te laai
    Stel die taakstatus op voltooi
    Keer die aflaaipad van F_out terug

    IF any error occurs THEN
        Set the job status to failed with the error message
    END IF

    INDIEN enige fout DAN voorkom
        Stel die taakstatus op misluk met die
    foutboodskap END IF

    The client periodically checks status_url
    IF the job status is completed THEN
    Download the translated document
    END IF
END

Die kliënt kontroleer periodiek status_url
    AS die posstatus DAN voltooi is
    Laai die vertaalde dokument af
    EINDE INDIEN
EINDE
```

The pseudocode describes the overall processing workflow of the proposed system. For DOCX documents, the system directly processes paragraphs, tables, headers, and

Die pseudokode beskryf die algehele verwerkingswerkvloei van die voorgestelde stelsel. Vir DOCX-dokumente verwerk die stelsel paragrawe, tabelle, opskrifte en footers to translate textual content while preserving the original formatting. For PDF documents, the system uses DOCX as an intermediate editable format by converting the PDF into DOCX, translating the extracted content, recovering the layout, and exporting the result back to PDF. During translation, the system supports three modes: standard translation, side-by-side bilingual translation, and line-by-line bilingual translation. In addition, special elements such as URLs, DOIs, citations, formulas, and technical symbols are protected to prevent incorrect translation of non-translatable content.

voettekste om tekstuele inhoud te vertaal terwyl die oorspronklike formatering behoue bly. Vir PDF-dokumente gebruik die stelsel DOCX as 'n intermediêre redigeerbare formaat deur die PDF in DOCX om te skakel, die onttrekte inhoud te vertaal, die uitleg te herstel en die resultaat terug na PDF uit te voer. Tydens vertaling ondersteun die stelsel drie modusse: standaardvertaling, sy-aan-sy tweetalige vertaling en reël-vir-lyn tweetalige vertaling. Daarbenewens word spesiale elemente soos URL's, DOI's, aanhalings, formules en tegniese simbole beskerm om verkeerde vertaling van nie-vertaalbare inhoud te voorkom.

After translation and layout recovery are completed, the system exports the output file according to the original document type. If the input file is DOCX, the system returns a translated DOCX file. If the input file is PDF, the system converts the translated DOCX back to PDF and performs output quality checking. This checking step aims to detect serious errors such as missing content, missing tables, missing images, or layout problems that affect readability. The final file is saved in the download directory, and the download path is returned to the user after the job status changes to completed.

Nadat vertaling en uitlegherwinning voltooi is, voer die stelsel die uitvoerlêer uit volgens die oorspronklike dokumenttipe. As die invoerlêer DOCX is, gee die stelsel 'n vertaalde DOCX-lêer terug. As die invoerlêer PDF is, skakel die stelsel die vertaalde DOCX terug na PDF en voer die kwaliteit kontrolering uit. Hierdie kontrolestap het ten doel om ernstige foute op te spoor soos ontbrekende inhoud, ontbrekende tabelle, ontbrekende beelde of uitlegprobleme wat leesbaarheid beïnvloed. Die finale lêer word in die aflaaigids gestoor, en die aflaaipad word aan die gebruiker teruggestuur nadat die taakstatus na voltooi verander het.

4 Experiments
4 Eksperimente

The experiments were conducted to evaluate the system from two main perspectives: formatting preservation after translation and translation content quality. Separating these two evaluation groups is necessary because document translation depends not only on linguistic quality but also on the ability to preserve the original visual structure of the document. A translation may be semantically correct but still fail in practice if it loses tables, changes the layout, or produces an unreadable output file. Conversely, a document may preserve its formatting well, but the translation may still be insufficient if it is unnatural or uses incorrect terminology.

Die eksperimente is uitgevoer om die stelsel vanuit twee hoofperspektiewe te evalueer: formateringbewaring na vertaling en vertalinginhoudkwaliteit. Die skeiding van hierdie twee evalueringsgroepe is nodig omdat dokumentvertaling nie net afhang van linguistiese kwaliteit nie, maar ook van die vermoë om die oorspronklike visuele struktuur van die dokument te bewaar. 'n Vertaling kan semanties korrek wees, maar misluk steeds in die praktyk as dit tabelle verloor, die uitleg verander of 'n onleesbare uitvoerlêer produseer. Omgekeerd kan 'n dokument sy formatering goed bewaar, maar die vertaling kan steeds onvoldoende wees as dit onnatuurlik is of verkeerde terminologie gebruik.

The experiments were conducted on an Acer Nitro AN515-45 computer running Microsoft Windows 11 Home Single Language 64-bit. The machine uses an AMD Ryzen 55600H with Radeon Graphics processor, with 12 logical processors at approximately 3.3 GHz, on an x64-based PC architecture. The system is equipped with 8 GB of RAM and DirectX 12. The development environment uses Python together with the FastAPI framework on the backend for receiving document translation requests, managing background processing jobs, and returning output files to users. The user interface is implemented using HTML, CSS, and JavaScript. For document processing, the system uses pdf2docx for PDF-to-DOCX conversion, DOCX processing libraries for handling paragraphs, tables, headers, and footers, Tesseract OCR for scanned documents or embedded images, and a DOCX-to-PDF conversion tool for generating the final translated PDF output. The translation providers used in the experiments include Google Translate, DeepL Translator, and Gemini 2.5 Flash.

Die eksperimente is uitgevoer op 'n Acer Nitro AN515-45 rekenaar met Microsoft Windows 11 Home Single Language 64-bis. Die masjien gebruik 'n AMD Ryzen 55600H met Radeon Graphics-verwerker, met 12 logiese verwerkers teen ongeveer 3.3 GHz, op 'n x64-gebaseerde rekenaarargitektuur. Die stelsel is toegerus met 8 GB RAM en DirectX 12. Die ontwikkelingsomgewing gebruik Python saam met die FastAPI-raamwerk op die agterkant vir die ontvangs van dokumentvertalingsversoeke, die bestuur van agtergrondverwerkingstake en die terugstuur van uitvoerlêers aan gebruikers. Die gebruikerskoppelvlak word geïmplementeer met behulp van HTML, CSS en JavaScript. Vir dokumentverwerking gebruik die stelsel pdf2docx vir PDF-na-DOCX-omskakeling, DOCX-verwerkingsbiblioteke vir die hantering van paragrawe, tabelle, kop- en voettekste, Tesseract OCR vir geskandeerde dokumente of ingebedde beelde, en 'n DOCX-na-PDF-omskakelingsinstrument vir die generering van die finale vertaalde PDF-uitset. Die vertaalverskaffers wat in die eksperimente gebruik is, sluit in Google Translate, DeepL Translator en Gemini 2.5 Flash.

The experimental data consists of DOCX and PDF academic documents containing headings, paragraphs, tables, images, and page layouts. These documents were processed using three translation modes: standard translation, side-by-side bilingual translation, and line-by-line bilingual translation. For format evaluation, the system considers the following criteria: Heading, Text/Style, Table, Layout, and No

Die eksperimentele data bestaan uit DOCX en PDF akademiese dokumente wat opskrifte, paragrawe, tabelle, beelde en bladsyuitlegte bevat. Hierdie dokumente is verwerk met behulp van drie vertaalmodusse: standaardvertaling, tweetalige vertaling langs mekaar en tweetalige lyn-vir-lyn vertaling. Vir formaatevaluering neem die stelsel die volgende kriteria in ag: Opskrif, Teks/Styl, Tabel, Uitleg en Nee

Overflow. Heading measures whether titles and section hierarchy are preserved. Text/Style evaluates whether fonts, paragraphs, and textual presentation remain readable. Table measures whether table structures are preserved. Layout reflects the overall page organization. No Overflow measures whether the output document avoids serious text overflow or overlapping. Minor errors that can be manually corrected, such as slight line shifts or natural page increases caused by longer translated text, are not considered serious errors if they do not affect readability.

Oorloop. Opskrif meet of titels en afdelinghiërargie behoue bly. Teks/Styl evalueer of lettertipes, paragrawe en teksaanbieding leesbaar bly. Tabel meet of tabelstrukture behoue bly. Uitleg weerspieël die algehele bladsyorganisasie. Geen oorloop meet of die uitvoerdokument ernstige teksoorloop of oorvleueling vermy. Geringe foute wat met die hand reggestel kan word, soos effense lynverskuiwings of natuurlike bladsyverhogings wat veroorsaak word deur langer vertaalde teks, word nie as ernstige foute beskou as dit nie leesbaarheid beïnvloed nie.

The AVG score is calculated as the arithmetic mean of five formatting evaluation criteria: Heading, Text/Style, Table, Layout, and No Overflow. A higher AVG value indicates better formatting preservation and output readability.

Die AVG-telling word bereken as die rekenkundige gemiddelde van vyf formatering-evalueringskriteria: Opskrif, Teks/Styl, Tabel, Uitleg en Geen oorloop. 'n Hoër AVG-waarde dui op beter formateringbewaring en uitvoerleesbaarheid.

relative comparison among machine translation outputs rather than an absolute conclusion about translation quality.

relatiewe vergelyking tussen masjienvertalingsuitsette eerder as 'n absolute gevolgtrekking oor vertaalkwaliteit.

Overall, the experimental results show that the proposed system performs well for DOCX documents and achieves promising results for PDF documents. The difference between DOCX and PDF performance comes from the structural characteristics of each format. DOCX allows direct processing of text and formatting components, while PDF requires several intermediate conversion steps, which increases the possibility of formatting errors. For bilingual translation modes, document length and page count may increase because both the source and translated text are retained. This should not be considered a formatting error as long as the document remains readable, complete, and free from serious text overlapping.

Oor die algemeen toon die eksperimentele resultate dat die voorgestelde stelsel goed presteer vir DOCX-dokumente en belowende resultate vir PDF-dokumente behaal. Die verskil tussen DOCX- en PDF-prestasie kom van die strukturele kenmerke van elke formaat. DOCX laat direkte verwerking van teks- en formateringskomponente toe, terwyl PDF verskeie intermediêre omskakelingstappe vereis, wat die moontlikheid van formateringsfoute verhoog. Vir tweetalige vertaalmodusse kan dokumentlengte en bladsytelling toeneem omdat beide die bron- en vertaalde teks behoue bly. Dit moet nie as 'n formateringsfout beskou word nie solank die dokument leesbaar, volledig en vry van ernstige teksoorvleueling bly.

Despite the obtained results, the system still has several limitations that need further improvement. The PDF→DOCX→translation→PDF pipeline cannot guarantee perfect formatting for all types of PDF documents, especially documents with multi-column layouts, complex tables, mathematical formulas, or professional desktop-publishing designs. In addition, scanned PDFs depend heavily on OCR quality; if images are blurred, skewed, noisy, or use unusual fonts, OCR errors may affect translation quality. Since BLEU in this study is used in a cross-output setting due to the absence of a complete manually reviewed reference translation dataset, future work should construct a reference dataset and combine additional metrics such as METEOR, TER, and COMET for a more comprehensive evaluation(Banerjee& Lavie, 2005; Snover et al., 2006; Rei et al., 2020).

Ten spyte van die resultate wat verkry is, het die stelsel steeds verskeie beperkings wat verdere verbetering benodig. Die PDF→DOCX→vertaling→PDF-pyplyn kan nie perfekte formatering vir alle soorte PDF-dokumente waarborg nie, veral dokumente met multi-kolom-uitlegte, komplekse tabelle, wiskundige formules of professionele desktop-publisering-ontwerpe. Daarbenewens hang geskandeerde PDF's baie af van OCR-kwaliteit; as beelde vaag, skeef, raserig is of ongewone lettertipes gebruik, kan OCR-foute die vertaalkwaliteit beïnvloed. Aangesien BLEU in hierdie studie in 'n kruisuitsetopstelling gebruik word as gevolg van die afwesigheid van 'n volledige handmatig hersiene verwysingsvertalingsdatastel, behoort toekomstige werk 'n verwysingsdatastel te konstrueer en addisionele maatstawwe soos METEOR, TER en COMET te kombineer vir 'n meer omvattende evaluering (Banerjee & Lavie, 2005; Snover et al.06, 20).

5 Conclusion

5 Gevolgtrekking

This study proposed and developed an AI-based document translation system with original format preservation for two main document formats: DOCX and PDF. The system supports three translation modes: standard translation, side-by-side bilingual translation, and line-by-line bilingual translation. It also allows users to select different translation providers, including Google Translate, DeepL Translator, and Gemini 2.5 Flash. For DOCX documents, the system directly processes structured components such as paragraphs, tables, headers, and footers. For PDF documents, the system uses a PDF→DOCX→translation→layout recovery→PDF pipeline to take advantage of DOCX editability and improve format preservation.
Hierdie studie het 'n KI-gebaseerde dokumentvertalingstelsel met oorspronklike formaatbewaring voorgestel en ontwikkel vir twee hoofdokumentformate: DOCX en PDF. Die stelsel ondersteun drie vertaalmodusse: standaardvertaling, sy-aan-sy tweetalige vertaling en reël-vir-lyn tweetalige vertaling. Dit stel gebruikers ook in staat om verskillende vertaalverskaffers te kies, insluitend Google Translate, DeepL Translator en Gemini 2.5 Flash. Vir DOCX-dokumente verwerk die stelsel gestruktureerde komponente soos paragrawe, tabelle, kop- en voettekste direk. Vir PDF-dokumente gebruik die stelsel 'n PDF→DOCX→vertaling→uitlegherstel→PDF-pyplyn om voordeel te trek uit DOCX-redigeerbaarheid en formaatbewaring te verbeter.

The experimental results show that the system achieves very strong format preservation for Word documents and promising results for PDF documents. The bilingual translation modes are useful for helping users compare the original content with the translated content, especially in learning, research, and terminology checking. In terms of translation quality, BLEU Mean and BLEU Std show that the translation providers achieved relatively close results on the test dataset, with Gemini 2.5 Flash obtaining the highest score in the cross-output BLEU experiment. However, to evaluate translation quality more accurately, future research should construct manually reviewed reference translations and combine additional evaluation metrics.

Die eksperimentele resultate toon dat die stelsel baie sterk formaatbewaring vir Word-dokumente behaal en belowende resultate vir PDF-dokumente. Die tweetalige vertaalmodusse is nuttig om gebruikers te help om die oorspronklike inhoud met die vertaalde inhoud te vergelyk, veral in leer, navorsing en terminologiekontrole. Wat vertalingskwaliteit betref, toon BLEU Mean en BLEU Std dat die vertaalverskaffers relatief goeie resultate op die toetsdatastel behaal het, met Gemini 2.5 Flash wat die hoogste telling in die kruisuitset BLEU-eksperiment behaal het. Om vertalingskwaliteit meer akkuraat te evalueer, moet toekomstige navorsing egter handmatig hersiene verwysingsvertalings saamstel en bykomende evalueringsmaatstawwe kombineer.

In future work, the system can be improved in several directions. First, PDF processing can be enhanced by improving layout analysis, table recognition, and post-translation layout recovery. Second, OCR quality should be improved for scanned

In toekomstige werk kan die stelsel in verskeie rigtings verbeter word. Eerstens kan PDF-verwerking verbeter word deur uitleganalise, tabelherkenning en na-vertaling uitlegherwinning te verbeter. Tweedens, OCR-kwaliteit moet verbeter word vir geskandeer

PDFs and embedded images. Third, a domain-specific glossary can be added to
PDF's en ingebedde beelde. Derdens kan 'n domeinspesifieke woordelys bygevoeg word

ensure terminology consistency in academic and technical documents. Finally,
verseker terminologiekonsekwentheid in akademiese en tegniese dokumente. Uiteindelik,

support for additional formats such as PPTX, XLSX, and HTML can be developed to
ondersteuning vir bykomende formate soos PPTX, XLSX en HTML kan ontwikkel word

broaden the practical applicability of the system.
die praktiese toepaslikheid van die stelsel te verbreed.

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